Part A Dictionaries

```
inventory = {}
```

```
# 2. Store the first product details in variables
ProductNo1_name = "Mobile Phone"
ProductNo1_quantity = 5
ProductNo1_price = 20000
ProductNo1_releaseYear = 2020
```

```
# 3. Add first product details to inventory
inventory["ProductNo1"] = {
    "Product Name": ProductNo1_name,
    "Product Quantity": ProductNo1_quantity,
    "Product Price": ProductNo1_price,
    "Product Release Year": ProductNo1_releaseYear
}
```

```
# 4. Store the second product details in variables
ProductNo2_name = "Laptop"
ProductNo2_quantity = 3
ProductNo2_price = 50000
ProductNo2_releaseYear = 2022
```

```
# 5. Add second product details to inventory
inventory["ProductNo2"] = {
    "Product Name": ProductNo2_name,
    "Product Quantity": ProductNo2_quantity,
    "Product Price": ProductNo2_price,
    "Product Release Year": ProductNo2_releaseYear
}
```

```
# 6. Display the products present in inventory
print("Products in Inventory:")
print(inventory)
```

```
Products in Inventory:
{'ProductNo1': {'Product Name': 'Mobile Phone', 'Product Quantity': 5, 'Product Price': 20000, 'Product Release Year': 2020}, 'ProductNo2': {'Product Name': 'Laptop', '
```

```
# 7. Check if release years are in inventory
print(ProductNo1_releaseYear in inventory)
```

```
print(ProductNo2_releaseYear in inventory)
```

```
False
False
```

```
# 8. Delete release year of both products
del inventory["ProductNo1"]["Product Release Year"]
del inventory["ProductNo2"]["Product Release Year"]
```

```
print("Inventory after deleting release years:")
print(inventory)
```

```
Inventory after deleting release years:
{'ProductNo1': {'Product Name': 'Mobile Phone', 'Product Quantity': 5, 'Product Price': 20000}, 'ProductNo2': {'Product Name': 'Laptop', 'Product Quantity': 3, 'Product
```

Part B

```
# Create a tuple of product prices
prices = (250, 300, 150, 400, 100, 350, 200)
```

```
# Find highest and lowest price
print("Highest price:", max(prices))
print("Lowest price:", min(prices))
```

```
Highest price: 400
Lowest price: 100
```

```
# Calculate total sum
print("Total sum:", sum(prices))
```

```
Total sum: 1750
```

```
# Convert tuple into sorted list
sorted_prices = sorted(prices)
print("Sorted prices:", sorted_prices)
```

```
Sorted prices: [100, 150, 200, 250, 300, 350, 400]
```

```
# Try to modify an element
prices[0] = 500
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_1549/173560666.py in <cell line: 0>()
      1 # Try to modify an element
----> 2 prices[0] = 500

TypeError: 'tuple' object does not support item assignment
```

Why?

A tuple is immutable, which means we cannot change, add, or delete its elements after creating it.

```
prices = [250, 300, 150]
prices[0] = 500
```